

COMPUTER VISION AND AI-BASED CHILD SAFETY MONITORING FOR
MONTESSORI ENVIRONMENTS

Research Project Proposal
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Computer Science

by

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1.Introduction

1.1 Background of the Study

Unintentional injuries are the leading cause of death for children ages 1 to 14 globally with approximately 830,000 child deaths per year according to the WHO. Childcare centers report injury rates of 11 to 18 injuries per 100 children per year, with falls accounting for nearly half.

Montessori classrooms create a unique safety challenge. Children aged 2.5 to 6 move freely, choose their own activities and work independently with minimal direct supervision. The teacher acts as an observer rather than a direct supervisor. This freedom while educationally valuable makes it difficult to detect dangerous situations in real time.

Five dangerous activities are most common in these settings - falling, climbing on furniture or shelves, fighting between children, leaving the classroom unsupervised and entering restricted zones such as staircases , kitchen or storage areas.

Traditional CCTV monitoring depends on human operators, but research shows operators miss up to 95% of activity after 20 minutes due to attention fatigue. Recent advances in computer vision now make automated detection feasible. Object detection models like YOLO can track people at 30+ FPS and pose estimation tools like MediaPipe can extract body skeleton data in real time. However, no existing system applies these technologies to detect dangerous child activities in Montessori environments.

1.2 Motivation

Childhood injury statistics are alarming 75% of daycare injuries are preventable and most occur during free play which is the core of Montessori education. Human supervision is limited by staff-to-child ratios (typically 1:10 for preschoolers), attention fatigue and blind spots. The five target activities (falls, climbing, fighting, leaving, entering danger zones) are rapid events that human observers are most likely to miss.

While AI monitoring systems exist for elderly fall detection and industrial safety, no system has been designed for detecting dangerous activities among freely moving children in educational settings. This project addresses that gap.

1.3 Study Objectives

Primary Objective

- To design, develop and evaluate a computer vision and AI-based safety monitoring system that detects five specific categories of dangerous child activities falling, climbing, fighting, leaving the classroom and entering restricted danger zones in Montessori classroom environments in real time.

Secondary Objectives

- To implement a person detection and tracking module using YOLO that reliably detects and tracks children in classroom video feeds.
- To develop a pose estimation-based activity classification module using MediaPipe that distinguishes between the five dangerous activities and normal safe activities.
- To create a zone-based boundary monitoring module that detects classroom exit events and restricted area incursions using virtual boundary polygons.
- To design a real-time alert system that notifies teachers within seconds of detecting a dangerous activity, with information about the activity type and location.
- To evaluate the system's classification performance across all activity types using precision, recall, F1-score, and confusion matrix analysis.
- To implement privacy-preserving approaches such as edge computing and pose-only processing to protect children's data.

2. Literature Review and Research Gap

2.1 Overview of Existing Studies

Study 1: Toddler Fall Detection Using Improved YOLOv8 (2024) , Published in Sensors (MDPI, 24(19), 6451). Used improved YOLOv8 with GELAN for fall detection in toddlers aged 13-30 months, tested on 230 videos. Limitation- detects only falls, not other dangerous activities.

Study 2: Pose-Based Fall Detection (2022) , Salimi et al. in Sensors (22(12), 4544). Used 18 body keypoints with 1D - CNN to achieve 98% fall detection accuracy. Proved that skeleton data alone is sufficient for fall detection. Limitation- tested on adults only.

Study 3: MediaPipe Fall Detection (2025) - Achieved 100% accuracy on the UR - Fall Dataset using MediaPipe pose landmarks with threshold based rules. Confirms MediaPipe's effectiveness for fall detection.

Study 4: Hybrid YOLOv8s + AlphaPose (2025) - Published in Scientific Reports (Nature). Combined YOLO for person detection with AlphaPose for pose estimation improving accuracy by 4.30% and FPS by 37.50%. Validates the YOLO + pose estimation pipeline used in this project.

Study 5: AI Child Behavior Monitoring (2026) - Mahor et al. via IGI Global. CNN - LSTM system achieving 94.5% accuracy for classifying children's normal vs. suspicious activities. One of the few child specific studies but limited activity categories.

Study 6: Violence Detection Using RWF-2000 – Pose based fight detection using LSTM/ST - GCN on 2,000 surveillance clips achieves 82 - 94% accuracy. Designed for adults but the detection logic is adaptable to children.

Study 7: CV Tasks for Children's Health (2023) - Survey in Information (MDPI, 14(10), 548) found that most CV systems are trained only on adults, confirming a critical lack of child specific models.

2.2 Identification of Research Gap

- No system detects multiple dangerous activities (fall, climb, fight, leaving, zone entry) within a single framework for children.
- No monitoring system exists specifically for Montessori or child- centered educational environments.
- Most activity recognition models are built and tested only on adults not children.
- The YOLO + MediaPipe pipeline has not been applied to multi activity child safety detection.
- No privacy preserving activity monitoring framework exists for children in educational settings.

This study addresses all five gaps by building the first integrated, multi-activity detection system designed for Montessori classrooms.

3. Research Problem and Questions

3.1 Main Research Problem

Montessori classrooms experience recurring dangerous child activities falling, climbing, fighting, unsupervised classroom exits and entry into restricted zones that human observers frequently fail to detect in real time due to attention fatigue, blind spots and limited staff to child ratios. No existing computer vision or AI-based system has been designed to detect these specific dangerous activities in environments where children aged 2 to 6 move freely with minimal direct supervision.

3.2 Specific Research Questions

1. How accurately can a pose estimation based system using MediaPipe distinguish between the three pose dependent dangerous activities (falling, climbing and fighting) and normal safe activities among children in a Montessori style environment?
2. How effectively can YOLO based person detection combined with virtual boundary polygon logic detect the two zone based dangerous activities (leaving the classroom and entering restricted danger zones) in real time?
3. Can an integrated system combining pose based activity classification and zone based boundary monitoring achieve real time performance (≥ 15 FPS) while maintaining safety critical recall rates ($\geq 90\%$) across all five dangerous activity types?
4. What is the system's classification performance across all six activity classes (five dangerous + one normal/safe) as measured by the confusion matrix and which activity types present the greatest detection challenges?

4. Research Strategy and Methodology

4.1 Research Strategy

This project uses Design Science Research (DSR) which focuses on creating and evaluating IT artifacts to solve identified problems (Hevner et al., 2004, MIS Quarterly). DSR is suitable because the project outcome is a working software system that must be built, tested and evaluated against defined criteria.

4.2 Research Methodology Framework

The project follows Peffers' DSRM (2007), a six-phase process:

1. **Problem Identification** - Dangerous child activities go undetected in Montessori classrooms due to human supervision limitations.
2. **Define Objectives** - Recall $\geq 90\%$, precision $\geq 85\%$, ≥ 15 FPS, alert latency < 5 seconds, privacy-preserving design.
3. **Design and Development** - Build three modules: YOLO person detection, MediaPipe activity classification and zone-based boundary monitoring with real-time alerts.
4. **Demonstration** - Test using recorded video from a simulated Montessori environment with scripted dangerous scenarios.
5. **Evaluation** - Measure per-class precision, recall, F1-score generate confusion matrix measure FPS and latency.
6. **Communication** - Document findings in the undergraduate dissertation.

4.3 Data Collection Methods

Public datasets -

- Person detection: MS COCO pre-trained YOLO weights
- Fall detection: URFD (70 sequences), Le2i (221 videos), UP-Fall, CAUCAFall datasets
- Fight detection: RWF-2000 (2,000 clips), Hockey Fight Dataset (1,000 clips)
- Climbing: Kinetics-400 climbing action classes + rule-based pose geometry
- Pose estimation: MediaPipe
- Zone detection: No training data needed

Transfer learning - YOLO pre-trained on COCO detects persons directly. MediaPipe generalizes across body types and ages. The LSTM/1D-CNN activity classifier is trained on skeleton sequences extracted from the above public datasets. Pose normalization makes the classifier scale invariant, allowing adult trained models to generalize to children.

Simulated evaluation environment - A room set up to replicate a Montessori classroom with low shelving, child-sized furniture, exit points and restricted zones. Adult volunteers enact 30+ scripted instances per activity class (fall, climb, fight, leaving, entering danger zone, normal) recorded from ceiling mounted cameras.

4.4 Data Preprocessing / Preparation

- Frame extraction using OpenCV with 1–2 second sliding windows and 50% overlap
- MediaPipe extracts 33 body landmarks (132 features) per person per frame
- Pose landmarks normalized relative to bounding box for scale invariance
- Temporal sequences organized into fixed-length windows for LSTM input
- Augmentation: temporal jittering, Gaussian noise on keypoints, skeleton scaling, horizontal flipping

4.5 Model Development / Implementation

Module 1 - Person Detection: YOLO11 nano with COCO pre-trained weights (2.6M parameters). ByteTrack for multi-person tracking with persistent IDs.

Module 2 - Activity Classification: MediaPipe extracts pose per tracked person. Features computed per frame: body aspect ratio, vertical velocity, head-ankle distance (falls); feet elevation, upward displacement (climbing) inter person distance, limb velocity (fighting). A lightweight LSTM or 1D-CNN classifies temporal windows into (fall, climb, fight, normal). Parallel rule based thresholds provide a safety net for fall and climbing detection.

Module 3 - Zone Monitoring: Virtual polygons defined for exits and danger zones. Alerts trigger when a tracked person's centroid crosses an exit boundary or enters a restricted polygon (OpenCV pointPolygonTest). Temporal buffer of 5-10 frames prevents false triggers.

Alert System: Alerts include activity type, confidence, timestamp and annotated frame. Delivered via audible alarm, dashboard notification (WebSocket).

Pipeline: Camera → YOLO detection → ByteTrack → MediaPipe pose → feature computation → LSTM classifier + rule-based detector → zone checks → temporal validation → alert. Target: ≥ 15 FPS on edge hardware (NVIDIA Jetson Orin Nano).

4.6 Evaluation

- Per-class precision, recall and F1-score for all six classes (5 dangerous + 1 normal)
- 6×6 confusion matrix as the primary evaluation artifact
- Overall accuracy and macro averaged F1-score
- Real-time FPS (target ≥ 15) and detection-to-alert latency (target < 5 seconds)
- Per-class precision-recall curves for threshold optimization

4.7 Ethical considerations

- All processing on device (edge computing), no raw video stored or transmitted
- Only anonymous skeleton data (33 keypoints) and alert metadata retained
- Face anonymization applied by default aligning with GDPR data minimization (Article 5) and COPPA requirements
- Project uses only public datasets under their licenses and adult volunteers with written consent
- No real children recorded or involved at any stage
- All test data encrypted (AES-256) and deleted upon project completion

4.8 Limitations of the study

- Public datasets feature adults- children's different body proportions and movement patterns may reduce accuracy
- Simulated environment cannot fully replicate real classroom unpredictability
- Activity ambiguity - rough play vs. fighting, intentional lying down vs. falling
- Occlusion from multiple children in close proximity degrades detection
- System detects only five predefined activities; other hazards (choking, burns) are out of scope
- Excessive false alarms may cause alert fatigue among teachers

5. Timeline

01 November, 2025 To 01 November, 2026